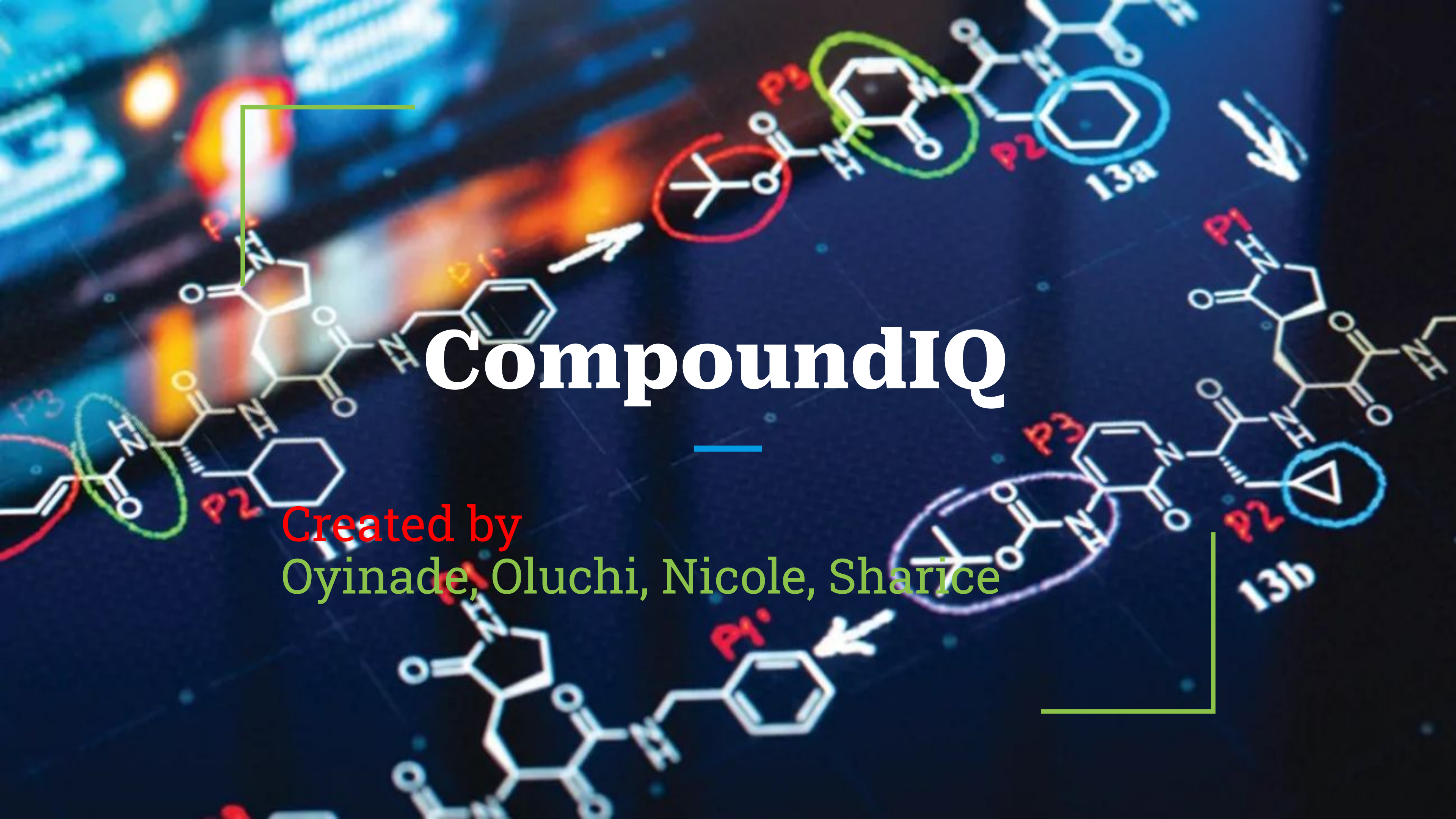

CompoundIQ

Created by

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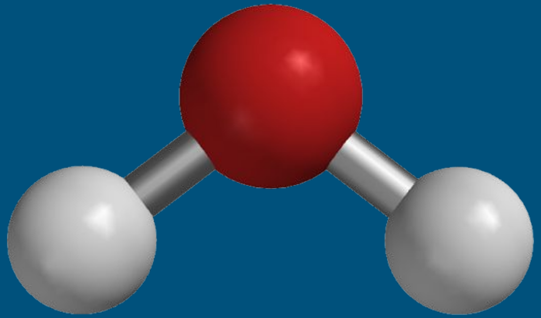
What Is CompoundIQ

- Our project is an AI system that helps students doing research choose which drug compounds to test first.

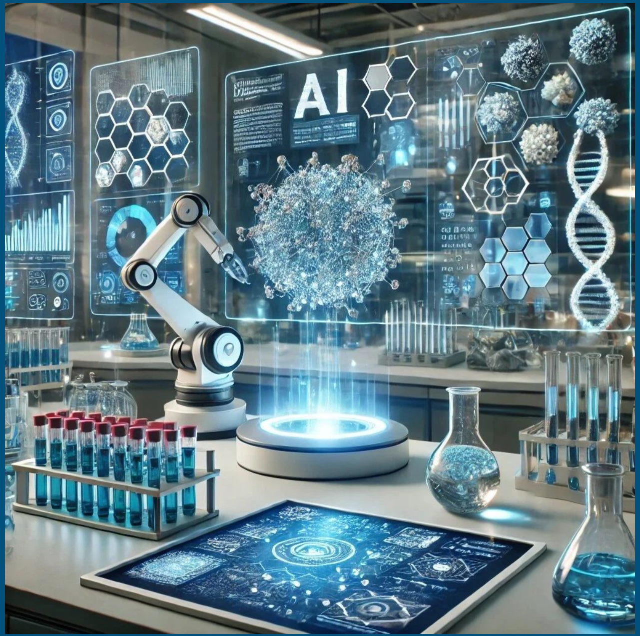
Goals for CompoundIQ

- **Provides hands-on learning experience for AI students studying molecular machine learning**
- **Makes predictions interpretable through attention visualization and feature attribution**



Goals for CompoundIQ

- **Offers basic virtual screening functionality to resource-constrained settings**
- **Serves as open-source foundation for further student research projects**



How It Works

- Uses **Graph Neural Networks (GNNs) with attention** to analyze molecular structures
- Predicts **ADMET properties**: absorption, distribution, metabolism, excretion, toxicity
- Helps researchers **prioritize compounds for lab testing**, reducing time and cost
- Explores **toxic substructure identification** via attention mechanisms
- Uses **multi-task learning** to predict multiple properties simultaneously

What It Does.

Three Main Modules

1. Property Predictor

- Analyzes molecular structure
- Predicts: Toxicity, Solubility, Blood-Brain Barrier penetration, Binding affinity, Bioavailability
- Shows attention heatmap highlighting important atoms (RED = critical, ORANGE = high, YELLOW = medium)
- Generates text explanation: "High toxicity because nitro group [42% attention] metabolizes to toxic compounds"

2. Compound Optimizer

- Takes existing drug + improvement goals
- Generates 500 structural variants automatically
- Predicts properties for all variants
- Ranks by multi-objective score
- Returns top 20 optimized molecules
- Example: "Reduce ibuprofen stomach damage 30% while maintaining anti-inflammatory activity"

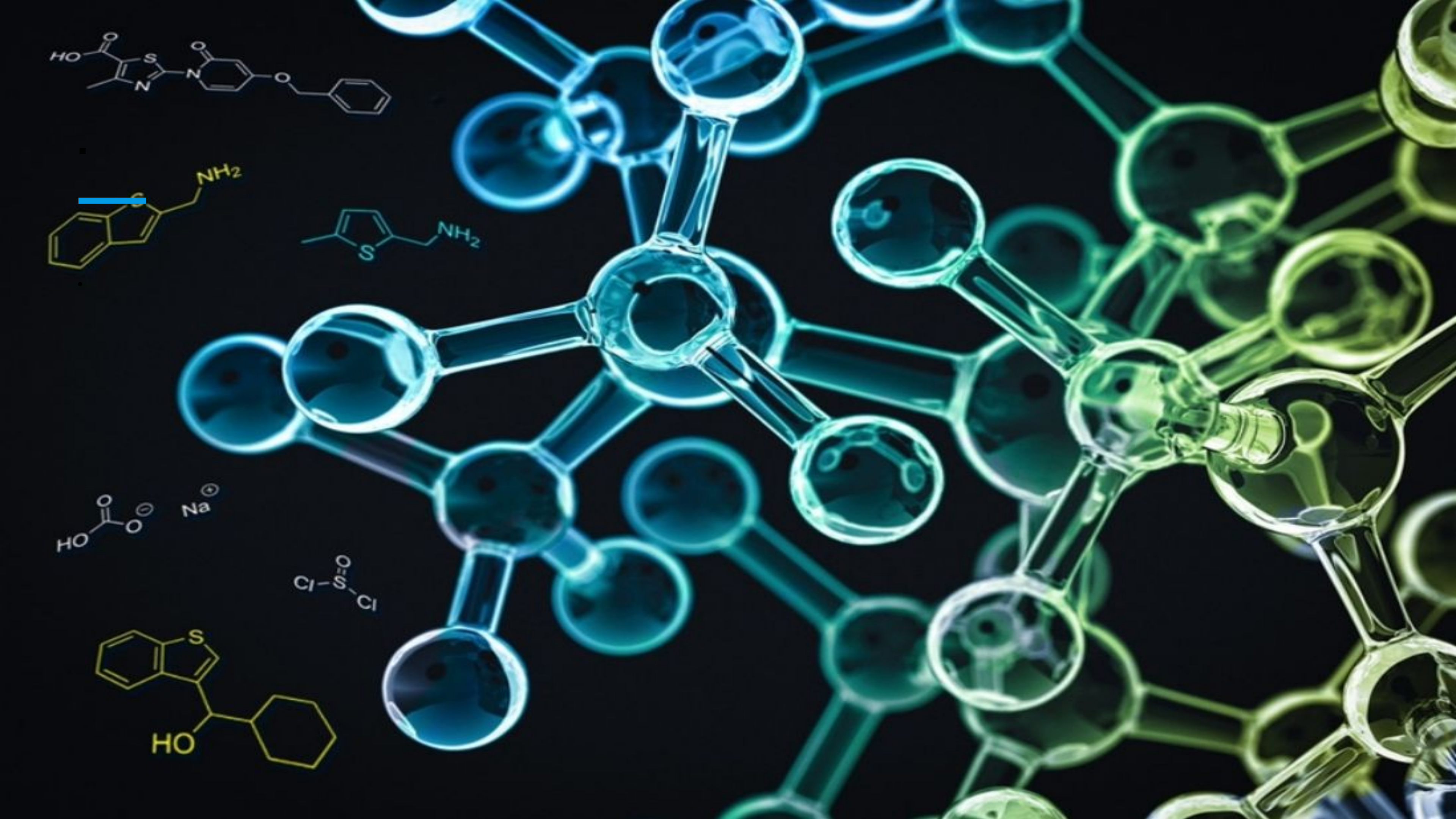
3. Virtual Screener

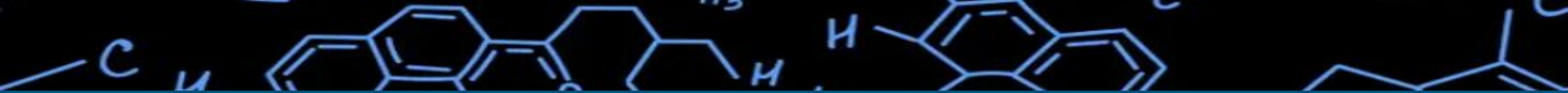
- Searches large molecular databases (10,000+ compounds)
- Filters for drug-likeness
- Predicts properties for all candidates
- Ranks by target disease profile
- Returns top 50 candidates with detailed predictions
- Example: "Find GABA receptor agonists with low sedation for muscle relaxants"

Technical Approach & Technologies

- **Graph Neural Networks (GNNs) & MPNNs** → analyze molecules as graphs
- **Attention Mechanisms** → highlight important atoms/bonds
- **Multi-task Learning** → predict multiple ADMET properties at once
- **Transfer Learning** → fine-tune pre-trained molecular embeddings

- **Compound Optimizer** → generate 100–500 variants, rank top candidates
- **Virtual Screening** → query ChEMBL/PubChem subsets, apply Lipinski filters
- **Interpretability** → attention heatmaps, SHAP feature attribution
- **Development Tools** → Python, PyTorch, RDKit, NetworkX, Streamlit (all Colab-ready)





Molecular Databases for CompoundIQ

Primary Databases (Required - 4 databases)

ChEMBL - 2.4M bioactive compounds | https://www.ebi.ac.uk/chembl/ | Free API | Main training dataset (use 50K)

Tox21 - 12K compounds, 12 toxicity assays | https://tripod.nih.gov/tox21/ | Free download | Toxicity benchmark

ESOL - 1.1K compounds with solubility data | DeepChem library | Free | Solubility prediction

BBBP - 2K compounds with BBB labels | DeepChem library | Free | Blood-brain barrier prediction

Secondary Databases (Optional)

PubChem - 111M compounds | https://pubchem.ncbi.nlm.nih.gov/ | Free API | Virtual screening

ZINC15 - 230M purchasable compounds | https://zinc15.docking.org/ | Free subsets | Virtual screening

BindingDB - 2.8M binding datapoints | https://www.bindingdb.org/ | Free | Binding affinity

DrugBank - 14K drugs | https://go.drugbank.com/ | Free academic | Reference/validation

CARD - Antibiotic resistance data | https://card.mcmaster.ca/ | Free | Antibacterial use cases

Project Focus

Three gaps persist in computational drug discovery: (1) **Educational Access** - commercial platforms cost $10K-$100K annually, making them unavailable to community college students who can study theory but not practice implementation; (2) **Interpretability** - existing tools predict properties but don't explain why certain molecular structures cause specific results, reducing trust and learning opportunities; (3) **Resource Constraints** - small academic labs and developing region researchers cannot afford licensing costs, creating unequal access to computational chemistry tools.

CompoundIQ addresses these gaps by building an educational, interpretable, accessible platform that:

1. Provides hands-on learning experience for AI students studying molecular machine learning
2. Makes predictions interpretable through attention visualization and feature attribution
3. Offers basic virtual screening functionality to resource-constrained settings
4. Serves as open-source foundation for further student research projects

Timeline and Milestones

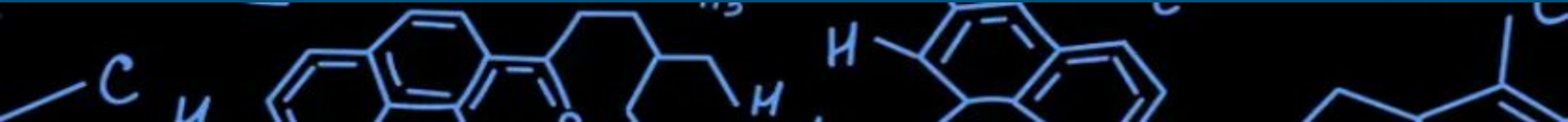
Week 1–2: Dataset prep & environment setup

Week 3–5: Build & test GNN prediction engine

Week 6–7: Add interpretability (attention heatmaps)

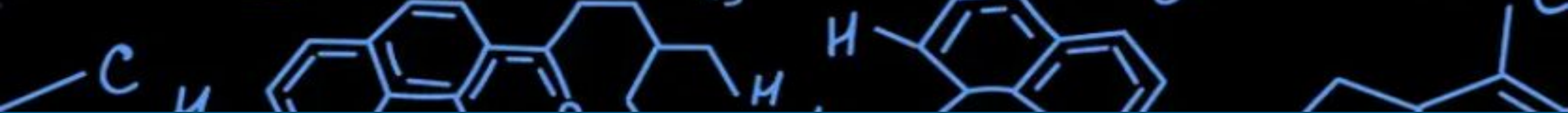
Week 8–9: Implement compound optimizer & screening

Week 10–12: Integration, testing, and presentation prep



Success Metrics

- Prediction accuracy for ADMET properties
- Correct ranking of top candidate compounds
- Quality of interpretability (heatmaps + explanations)
- System runs efficiently on test datasets



Risk Assessment

- **Data issues:** noisy or incomplete molecular datasets
- **Model complexity:** training GNNs may be slow or resource-heavy
- **Integration bugs:** combining modules could cause errors
- **Mitigation:** start small, use subsets, pre-trained models, test iteratively

Ethical considerations

- System is **educational & research support**, not a replacement for lab testing
- Predictions are **guidance only**, not medical advice
- Open-source approach promotes **equitable access**
- Transparency in AI reasoning reduces misuse or misinterpretation